

# Frontend – Assignment 1 (Short Answers)

## 1. HTTP Request–Response Cycle

The browser sends a request to the web server, and the server sends back a response. Steps:

- DNS lookup converts the domain name into an IP address.
- Browser connects to the server (TCP/TLS handshake).
- Browser sends an HTTP request (GET, POST, etc.).
- Server processes the request and prepares a response.
- Server sends back an HTTP response (status + data).
- Browser renders the received content.

## 2. Web Technologies

- HTML – defines the structure/content of a web page.
- CSS – styles and controls the look of the page.
- JavaScript – adds interactivity and dynamic behaviour.
- Bootstrap – CSS framework with ready-made responsive components.
- jQuery – JS library that simplifies DOM handling and AJAX.
- React.js – JS library for building UIs using reusable components.

## 3. Client-Side vs Server-Side

Client-side code (HTML, CSS, JS) runs in the browser and controls what the user sees. Server-side code (Node.js, PHP, Python, etc.) runs on the server and handles data, logic, and databases. Client-side is fast but visible/less secure; server-side is hidden and handles core logic.

## 4. Block-Level Elements

Elements that start on a new line and take up the full width available. Examples: <div>, <p>, <h1>–<h6>, <ul>, <table>.

## 5. Semantic Elements

Tags that describe their meaning (not just appearance), improving accessibility, SEO, and readability. Examples: <header>, <nav>, <article>, <section>, <footer>.

## 6. HTML Table

```
<table border="1">
  <tr><th>Sr. No.</th><th>Name</th><th>Designation</th><th>Department</th></tr>
  <tr><td>1</td><td>John</td><td>Manager</td><td>Sales</td></tr>
  <tr><td>2</td><td>Smith</td><td>Sr. Manager</td><td>Marketing</td></tr>
  <tr><td>3</td><td>Jane</td><td>Manager</td><td>HR</td></tr>
</table>
```

## 7. Nested List

```
<ul>
  <li>River
    <ul>
      <li>Ganga</li>
      <li>Yamuna</li>
      <li>Brahmaputra</li>
      <li>Narmada</li>
    </ul>
  </li>
</ul>
```

## 8. Types of CSS

- Inline CSS – style written directly in an element's style attribute.
- Internal CSS – written inside a <style> tag in the <head>.
- External CSS – written in a separate .css file and linked with <link>.

## 9. HTML vs CSS vs JavaScript

- HTML – structure content (the skeleton).
- CSS – presentation styling (the appearance).
- JavaScript – behaviour interactivity (the functionality).

## 10. Short Answers

- a. Bootstrap – gives ready-made, responsive UI components to speed up design.
- b. jQuery – JS library that simplifies DOM manipulation and AJAX; popular for cross-browser ease.
- c. React.js – JS library using components + Virtual DOM to build fast, interactive UIs.